

# Juan David Robles Fajardo

BOGOTÁ, COLOMBIA • UNIVERSIDAD NACIONAL DE COLOMBIA



CV-2026

## 01 PROFESSIONAL SUMMARY

Statistics undergraduate at *Universidad Nacional de Colombia*. Strong proficiency in Python, C++ and statistical modeling. Diverse experience in research-oriented projects including machine learning and Music Information Retrieval (MIR). Experience building predictive models and end-to-end data pipelines, from data collection and preprocessing to modeling and visualization. Fast self-learner with strong analytical and problem-solving skills.

## 02 FEATURED PROJECTS

### Phasing Generator

[GITHUB](#) →

Developed a stochastic algorithm generating Piano Phase-style musical sequences inspired by Steve Reich, using a custom heuristic scoring function based on psychoacoustic and music-theoretical principles. Used Monte Carlo simulation to explore score distributions and convergence behavior.

### Sound Identity (ongoing)

[GITHUB](#) →

Research project on automatic music classification and sound identity analysis using Machine Learning and Music Information Retrieval (MIR). Built a pipeline including web scraping, audio preprocessing, feature extraction, statistical filtering, and Random Forest classification.

## 03 WORK EXPERIENCE

### Private Mathematics Tutor

*Universidad Nacional de Colombia*

NOVEMBER 2024 -- PRESENT

*Bogotá, Colombia*

- Provide one-on-one tutoring to undergraduate students in mathematics and statistics.
- Teach mathematical foundations, number systems, linear algebra, and statistical inference.

### Python Instructor

*Kodland*

DECEMBER 2025 -- FEBRUARY 2026

*Remote*

- Taught introductory Python programming courses for children.
- Designed programming activities focused on computational thinking.

### Mathematics Tutor

*Independent*

JUNE 2025 -- DECEMBER 2025

*Bogotá, Colombia*

- Taught mathematics to high school students.

## 04 EDUCATION

### B.S. in Statistics (Ongoing)

*Universidad Nacional de Colombia*

JULY 2023 -- NOVEMBER 2027

*Bogotá, Colombia*

- Relevant coursework: Statistical Sampling, Regression Analysis, Statistical Inference, Matrix Algebra.

## 05 AWARDS

### Expresa3 Competition

*Universidad Nacional de Colombia*

3RD PLACE OVERALL

- Developed a spatial predictive model in R to identify urban crime patterns.

## 06 SYSTEM SPECS & SKILLS

### [ TECH STACK ]

- **Languages:** Python, R, C++
- **Data Science:** Pandas, NumPy, Scikit-learn, SciPy
- **Databases:** SQL
- **Visualization:** Matplotlib, GeoDa
- **Tools:** Git, GitHub, Jupyter Notebook, Google Colab, LaTeX, Microsoft Excel

### [ COURSES ]

- **MIR & Audio:** Music Computation and Complexity (Santa Fe Institute), Sound Design (MIT), Live Electronics Performance Practices (MIT)
- **Statistics & Data Science:** Advanced Excel (UNAL), Writing Efficient R Code (DataCamp), Reshaping Data with tidyr (DataCamp)

## 07 LANGUAGES

[ SPANISH ] Native    [ ENGLISH ] Fluent